Elementary reductions for the two-powers squarefree variant of Erdős Problem #11

Research note

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Abstract

Erdős Problem #11 asks whether every sufficiently large integer n can be written as $n = k + 2^m$ with k squarefree. We study a relaxation, the *two-powers variant*: every odd $n > 1$ satisfies $n = k + 2^l + 2^m$ for some squarefree k and $l, m \geq 0$. This variant is not one of Erdős's numbered problems; it appears as the statement `erdos_11.variants.two_pow_two` in formalised-conjectures repositories, together with a remark, which we have not independently verified against a primary source, that Erdős thought a two-power relaxation of #11 "might be easy". We give a sieve reduction in which the dominant term is controlled *exactly and uniformly in* n by a new cancellation lemma at non-Wieferich primes (Lemma 3.1, the p-adic reduction underlying the square-sieve literature, here specialised to this sieve), close the boundary term completely (§4), reduce the remaining worst-case obstruction to a single, clean, n-independent multiplicity statement (Lemma 5.2, §5), and prove that the two-powers variant holds *unconditionally for almost all* n (Corollary 6.9, density $1 - O(1/(L \log L))$, via a second moment on $(L, L^2]$ and a first moment at a raised threshold on the wide range $(L^2, \sqrt{n/2}])$. For the *every-n* statement we give a conditional theorem (Theorem 5.9): it holds provided a single grand-sieve bound on the multiplicity distribution of $\{2^l + 2^m \bmod p^2\}$ (hypothesis (TB)), together with two softer structural inputs. Along the way we prove unconditionally that the relevant order sum $\sum_{p \in (L, L^2]} \lfloor L / \operatorname{ord}_p(2) \rfloor$ is $o(L^2)$ (Proposition 5.4) and that the Type A count satisfies $\sum M_p = O(L^{8/3} / \log L)$ via García–Voloch (Proposition 5.7), reducing the obstruction to the Type B tail. For that tail we show that it is closed by *circularity*: Bourgain–Chang gives the additive sup-norm unconditionally (Theorem 5.11), but the moment bound needed to convert it into (TB) is *equal* to the additive energy, hence to (TB) itself (Proposition 5.13). A function-field model (§8) locates the difficulty as *archimedean* – the failure of $\{2^j\}$ to be exactly Sidon, powers carrying where monomials do not – rather than a lack of cancellation, and identifies (M$''$) at $k = 2$ as a squarefree-value statement, a cousin of #11. The remaining gaps are stated honestly with the exact point where elementary counting fails (§7, §8). All are decoupled from two classical obstructions one might expect to inherit: the infinitude of non-Wieferich primes (only the two *known* Wieferich primes are ever excluded, an $O(1)$ cost), and Crocker's 1971 disproof of the *prime*-kernel analogue $n = p + 2^a + 2^b$, which does not transfer to a squarefree kernel.

Contents

1 Introduction

1.1 The problem and its variant

Erdős Problem #11 [1] asks whether every sufficiently large n can be written as $n = k + 2^m$ with k squarefree and $m \geq 0$. The problem is open. Partial and conditional results exist in the literature attributed (see the exact reference list of Kalinin [2], who states and uses them) to Granville–Soundararajan, Hercher, Crocker, and Platt–Trudgian; we rely on [2] for the precise statements and do not reproduce them here, since cross-checking each primary source individually is outside the scope of this note.

We study the following relaxation, which lets n be written as a squarefree kernel plus *two* powers of 2 instead of one:

> ***Two-powers variant.*** *For every odd integer $n > 1$ there exist $k, l, m \geq 0$ with k squarefree and $n = k + 2^l + 2^m$.*

(Restricting to odd n costs nothing: if n is even, $n = k + 2^l + 2^m$ with k odd forces $l = m = 0$ to be excluded or absorbed trivially by parity, and the even case reduces to a one-power-type statement on $n/2$-scale considerations that we do not pursue here; all of our results concern odd n, matching the Lean statement.) This statement is recorded as `erdos_11.variants.two_pow_two` in formalised-conjectures Lean repositories. It is, to our knowledge, unpublished and unproved; this note does not close it, but reduces it as far as we have been able to push it with elementary and semi-elementary tools.

1.2 Contributions

1. **Lemma K** (§3): an exact character-sum cancellation, $\sum_{x\in\langle 2\rangle} e_{p^2}(ax) = 0$ for every non-Wieferich prime $p \geq 3$ and every a with $p \nmid a$, where $\langle 2\rangle$ is the subgroup of $(\mathbb{Z}/p^2)^\times$ generated by 2. The vanishing mechanism is standard (the p-adic reduction underlying the square-sieve literature); the contribution is its exact specialisation to this sieve, which kills, *exactly and uniformly in* n, the dominant "complete period" term of every medium-sized prime with no equidistribution input.

2. A complete elementary closure of the boundary term left over by Lemma K (Proposition 4.2, §4): $O(L^2/\log L) = o(L^2)$, where $L = \lfloor \log_2 n\rfloor$.

3. A deterministic, n-independent reduction (Lemma 5.2, §5) *dissolving the worst-case-n obstruction* on the single dominant remaining range, to a clean additive multiplicity question.

4. **Type A, unconditionally** (Proposition 5.7): a bound of García–Voloch on additive representations in the subgroup $\langle 2\rangle$ mod p gives $\sum_{\text{Type A}} M_p = O(L^{8/3}/\log L)$ unconditionally – short of the target $o(L^2)$ by exactly $L^{2/3}$.

5. **Type B is closed by circularity** (§5.5): the Bourgain–Chang additive sup-norm bound holds unconditionally on our sums (Theorem 5.11), but the moment bound that would convert it into the Type B hypothesis (TB) equals the additive energy, i.e. (TB) *itself* (Proposition 5.13). Thus (TB) is a genuine open estimate, not a statement awaiting the right reference.

6. **A function-field model** (§8) proves the $\mathbb{F}_q[t]$ analogue of (TB) by a degree count (Proposition 8.1) and shows the mechanism does not transpose, diagnosing the obstruction over $\mathbb{Z}$ as *archimedean* – the failure of $\{2^j\}$ to be exactly Sidon.

7. An unconditional theorem (§6), via a second moment and Markov, for *almost all* n (density $1 - O(1/(L \log L))$).

8. **Machine verification**: the elementary lemmas underpinning the unconditional results are formally verified in Lean 4 against Mathlib (`Erdos11_verified.lean`, five lemmas, `0 sorry`).

9. A precise, honest statement of the two remaining gaps (§7, §8), each pinned to the exact point where brute-force counting fails and at which scale.

1.3 Decoupling from two classical obstructions

Independence from Wieferich-prime infinitude. Lemma K requires p to be non-Wieferich. It is open whether infinitely many Wieferich primes exist (equivalently, whether infinitely many primes fail to be non-Wieferich); only two are known, 1093 and 3511. Our reduction does *not* require the (open, and widely believed) statement that non-Wieferich primes have density 1, or even that there are infinitely many: it only ever needs to discard the *known* Wieferich primes

$\leq \sqrt{n}$, an $O(1)$-many exclusion contributing $O(L)$ to every estimate below, completely harmless. If new Wieferich primes are discovered, the argument is unaffected; if it turned out that a positive density of primes were Wieferich (which would be a sensational and unexpected result), Lemma K would have to be reproved for those primes by other means, but this is not on the table.

Independence from Crocker's disproof. Crocker [3] disproved the *prime*-kernel two-powers statement: there exist infinitely many n with $n \neq p + 2^a + 2^b$ for every prime p. His method is a covering-congruence argument using Fermat numbers $2^{2^s} + 1$: he exhibits residue classes modulo a suitable Fermat number in which $2^a + 2^b$ is forced into a fixed congruence class that makes $n - 2^a - 2^b$ composite for every choice of a, b in the relevant range. This disproof does *not* transfer to the squarefree-kernel variant studied here, for a structural reason: "composite" is forced by a congruence condition of density essentially 1 in the relevant residue classes (almost every integer in a Fermat-number residue class is composite), whereas "non-squarefree" is a much rarer condition (density $1 - 6/\pi^2 \approx 0.392$ overall, and in particular a positive proportion of any fixed congruence class is squarefree, by elementary sieve estimates). Crocker's congruence trick gives no control over squarefreeness at all. This is one reason the squarefree-kernel variant is plausible while its prime-kernel cousin is false.

1.4 A structural cousin: Erdős Problem #9

Erdős Problem #9 concerns $n = p+2^k+2^l$ with p *prime* (not squarefree) – the same surface shape (a fixed-size kernel condition plus two powers of 2) but a different, and harder, kernel condition. The toolkit that controls a squarefree kernel (counting square divisors via a sieve, as here) is structurally different from the toolkit needed for a prime kernel (which needs control of prime gaps / sieve lower bounds, a strictly harder problem). We do not address #9 here; we mention it only to locate #11's two-powers variant precisely among nearby open problems of the same shape.

2 Setup and notation

Throughout, n is odd, $n > 1$, and $L := \lfloor \log_2 n \rfloor$. We count *ordered* pairs $(l, m) \in \{0, \ldots, L\}^2$ (using ordered pairs only inflates every count by a harmless factor close to 2); there are $T := (L+1)^2$ of them. We discard the (negligibly many, $O(L)$) pairs with $2^l + 2^m \geq n$, since they cannot contribute a valid decomposition; this only shrinks the pool of candidate pairs and makes every bound below conservative.

Definition 2.1. For a prime $p \geq 3$,

$$N_p(n) := \#\{(l, m) \in \{0, \ldots, L\}^2 : \ p^2 \mid n - 2^l - 2^m\} = \#\{(l, m) : \ 2^l + 2^m \equiv n \pmod{p^2}\}.$$

Write $d_p := \mathrm{ord}_{p^2}(2)$, $e_p := \mathrm{ord}_p(2)$, $\langle 2 \rangle := \{2^l \bmod p^2\}$ (the cyclic subgroup of $(\mathbb{Z}/p^2)^\times$ generated by 2, of order d_p), and $H_p := \{2^l \bmod p\}$ (the cyclic subgroup of $(\mathbb{Z}/p)^\times$ generated by 2, of order e_p). A prime p is *Wieferich* if $2^{p-1} \equiv 1 \pmod{p^2}$; equivalently $d_p = e_p$ (no extra factor of p is gained from $\mathbb{Z}/p$ to $\mathbb{Z}/p^2$). Only two Wieferich primes are known: 1093 and 3511. For non-Wieferich p, $d_p = p \cdot e_p$.

Since $k = n - 2^l - 2^m$ is always odd, $p = 2$ never occurs in any of the sieve sums below.

Proposition 2.2 (Basic reduction)**.** *The number of pairs $(l, m) \in \{0, \ldots, L\}^2$ for which $k = n - 2^l - 2^m$ is* not *squarefree is at most $\sum_{3 \leq p \leq \sqrt{n}} N_p(n)$. If this sum is $< T$, then the two-powers variant holds for n.*

Proof. Union bound over primes p with $p^2 \mid k$ for some valid pair. $\square$

(Small n are not a concern: the variant has been checked directly, without exception, up to 5×10^7.)

Proposition 2.3 (Periodicity bound)**.** *For every prime $p \geq 3$, $N_p(n) \leq \dfrac{(L+1)^2}{d_p} + (L+1)$.*

Proof. $2^l \bmod p^2$ is d_p-periodic in l. For each fixed $l \leq L$, the equation $2^m \equiv n - 2^l \pmod{p^2}$ has at most $\lceil (L+1)/d_p \rceil \leq (L+1)/d_p + 1$ solutions $m \in \{0, \dots, L\}$. Summing over the $L+1$ values of l gives the claim. $\square$

2.1 The pivot constant

The reduction below needs $\sum_{p \geq 3} 1/d_p < \frac{1}{2}$. We record two facts: the precise numerical value, and an elementary proof that the underlying infinite series in fact converges (so that summing it over an arbitrarily large range of primes, as we will, is legitimate).

Proposition 2.4 (Convergence)**.** *For every prime $p \geq 3$, $e_p > \log_2 p - 1$. Consequently, for every non-Wieferich p, $d_p = p\, e_p > p(\log_2 p - 1)$, and $\sum_{p \geq 3} 1/d_p$ converges (the two Wieferich primes contribute two extra, harmless, finite terms).*

Proof. By definition $p \mid 2^{e_p} - 1 < 2^{e_p}$, so $p < 2^{e_p}$, i.e. $e_p > \log_2 p$; we used the slightly weaker $e_p > \log_2 p - 1$ to avoid worrying about the base case. For the convergence: by Mertens' classical theorem, $\sum_{p \leq x} 1/p = \log\log x + M + o(1)$; partial summation against the decreasing weight $1/(\log_2 p - 1)$ turns this into the classical fact that $\sum_p \frac{1}{p \log p}$ converges (the density of primes near t is $1/\log t$ by the prime number theorem, so the partial sums behave like $\int^x \frac{dt}{t(\log t)^2}$, convergent). Since $1/d_p < 1/(p(\log_2 p - 1)) = O(1/(p \log p))$ for non-Wieferich p, comparison gives convergence of $\sum_p 1/d_p$. $\square$

Proposition 2.5 (Numerical value)**.** *$\sum_{3 \leq p < 5000} 1/d_p = 0.320516\dots$, computed exactly via exact modular-order arithmetic (script `orders.py`), dominated by $p = 3$ ($1/6$), $p = 5$ ($1/20$), $p = 7$ ($1/21$). By Proposition 2.4 the tail $p \geq 5000$ is provably summable and numerically negligible (a crude comparison with $\sum_{p \geq 5000} 1/(p(\log_2 p - 1))$ already gives less than 3×10^{-4}); in particular*

$$S := \sum_{p \geq 3} \frac{1}{d_p} \;\approx\; 0.3205 \;<\; \frac{1}{2}.$$

Remark 2.6. For the application below we only ever need a *fixed* threshold z (we use $z = 5000$) with $\sum_{p<z} 1/d_p < \frac{1}{2}$ exactly, plus the fact that the tail $\sum_{p \geq z} 1/d_p \to 0$ as $z \to \infty$ – the latter is exactly the convergence statement of Proposition 2.4, not a numerical estimate.

3 Lemma K: exact cancellation at non-Wieferich primes

The vanishing mechanism below – a multiplicative subgroup whose order is divisible by p necessarily contains the kernel $K = 1 + p\mathbb{Z}/p^2\mathbb{Z}$ of the reduction $(\mathbb{Z}/p^2)^\times \to (\mathbb{Z}/p)^\times$, and an additive character summed over a union of K-cosets vanishes – is standard in the theory of exponential sums modulo prime powers (it is the elementary p-adic reduction underlying, e.g., Cochrane–Zheng). What is specific to this note is only its exact application to the present sieve; we state it as a lemma for that reason, not because the identity itself is new.

Lemma 3.1 (Lemma K). *Let $p \ge 3$ be a non-Wieferich prime. For every integer a with $p \nmid a$,*

$$S(a) := \sum_{x \in \langle 2 \rangle} e_{p^2}(ax) \;=\; 0, \qquad e_{p^2}(t) := e^{2\pi i t/p^2}.$$

Proof. Since p is non-Wieferich, $d_p = p\, e_p$, so $\langle 2 \rangle$ (cyclic of order d_p) contains the unique subgroup of order p of $(\mathbb{Z}/p^2)^\times$, namely the kernel $K = \{1 + jp : \; j = 0, \ldots, p-1\}$ of the reduction map $(\mathbb{Z}/p^2)^\times \to (\mathbb{Z}/p)^\times$. (Indeed $\langle 2 \rangle \to H_p$ is surjective with kernel $\langle 2 \rangle \cap K$; comparing orders $p\, e_p$ and e_p, this kernel has order p, hence equals K.) Partition $\langle 2 \rangle$ into cosets gK over a set of representatives g:

$$S(a) = \sum_g \sum_{j=0}^{p-1} e_{p^2}\big(ag(1+jp)\big) = \sum_g e_{p^2}(ag) \underbrace{\sum_{j=0}^{p-1} e_p(agj)}_{=0},$$

the inner sum being geometric with ratio $e_p(ag) \ne 1$ since $p \nmid a$ and $p \nmid g$ (as $g \in (\mathbb{Z}/p^2)^\times$). □

Remark 3.2 (Numerical verification). For $p = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 127$ (script `lemmaE2.py`), all non-Wieferich, the kernel K is exactly contained in $\langle 2 \rangle$ and the maximum of $|S(a)|$ over $p \nmid a$ is exactly 0, confirming the lemma. For the two known Wieferich primes 1093 and 3511, $d_p = e_p$ (no extra factor of p), $K \not\subseteq \langle 2 \rangle$, and $S(a) \ne 0$: the mechanism breaks exactly where the hypothesis fails, as expected.

3.1 Exact structure on a complete period

Proposition 3.3 (Exact period count). *Let p be non-Wieferich. For every residue n mod p^2,*

$$\#\big\{(l,m) \in \{0,\ldots,d_p-1\}^2 : \; 2^l + 2^m \equiv n \pmod{p^2}\big\} \;=\; p \cdot r_p(n), \qquad r_p(n) := \#\big\{(u,v) \in H_p^2 : \; u+v \equiv n \quad (\mathrm{mod}\ p$$

Proof sketch. By Lemma 3.1 (via Fourier inversion on $\langle 2 \rangle$, or directly): each solution $(u,v) \in H_p^2$ of $u + v \equiv n \pmod{p}$ lifts to exactly p pairs $(x,y) \in \langle 2 \rangle^2$ with $x + y \equiv n \pmod{p^2}$, namely the p choices of x in the K-coset of u (each forcing y's residue mod p to be v, hence y's K-coset, but *not* pinning y further – the count is exactly balanced by the geometric cancellation of Lemma 3.1). Verified by direct computation for $p = 3, 5, 7, 11, 13, 17, 19, 23$ (script `verify_structure.py`): the identity holds exactly for every residue n mod p^2. □

Since $r_p(n) \le |H_p| = e_p$ (each $u \in H_p$ admits at most one matching v), this gives, for non-Wieferich p with $d_p \le L + 1$ (so that $Q_p := \lfloor (L+1)/d_p \rfloor \ge 1$ complete periods fit in $\{0, \ldots, L\}$):

$$Q_p^2\, p\, r_p(n) \;\le\; Q_p^2\, p\, e_p \;=\; Q_p^2\, d_p \;\le\; \frac{(L+1)^2}{d_p}, \qquad \text{uniformly in } n.$$

Theorem 3.4 (Complete periods are uniformly $o(L^2)$). *For every odd $n > 1$,*

$$\sum_{\substack{p \text{ non-Wief.} \\ d_p \le L+1}} Q_p^2\, p\, r_p(n) \;\le\; (L+1)^2 \sum_{p \ge 3} \frac{1}{d_p} \;\le\; 0.3205\, (L+1)^2,$$

uniformly in n, with no equidistribution hypothesis whatsoever.

Proof. Immediate from the displayed bound above, summed over the (at most $\pi(L+1)$, see Proposition 4.1 below) primes with $d_p \le L + 1$, and Proposition 2.5. □

This is the main term of the whole sieve, and it is now controlled *exactly*, for every n, by an elementary cancellation – no equidistribution, no analytic input, no conjecture.

4 Closing the boundary term: Proposition R1

Theorem 3.4 controls the contribution of full periods. What remains, for primes with $d_p \le L+1$, is the boundary correction between a whole number of periods and the actual finite range $\{0, \dots, L\}$.

Proposition 4.1 (Localisation)**.** *If p is non-Wieferich and $d_p \le L+1$, then $p \le L+1$.*

Proof. $d_p = p\, e_p \ge p$ since $e_p \ge 1$. ☐

So the primes carrying a boundary term are confined to $p \le L+1$, at most $\pi(L+1)$ of them.

Proposition 4.2 (R1, elementary closure)**.** *For n odd, define*

$$R1(n) := \sum_{\substack{p \text{ non-Wief.} \\ d_p \le L+1}} \big(N_p(n) - Q_p^2\, p\, r_p(n)\big).$$

Then $R1(n) \;<\; 3(L+1)\,\pi(L+1) \;=\; O\!\left(\dfrac{L^2}{\log L}\right) = o(L^2).$

Proof. Fix such a p, and write $L+1 = Q_p d_p + s_p$, $0 \le s_p < d_p$. On the range $\{0, \dots, L\}$, each residue $x \in \langle 2 \rangle$ is hit $c(x) = Q_p + \varepsilon(x)$ times, $\varepsilon(x) \in \{0,1\}$, with $\#\{x : \varepsilon(x) = 1\} = s_p$ (the residues $2^0, \dots, 2^{s_p-1}$). Writing $P_n := \{x \in \langle 2 \rangle : n - x \in \langle 2 \rangle\}$ (so $|P_n| = p\, r_p(n)$ by Proposition 3.3),

$$N_p(n) = \sum_{x \in P_n} c(x)c(n-x) = Q_p^2 |P_n| + 2Q_p \gamma_p(n) + \delta_p(n),$$

with $\gamma_p(n) := \#\{x \in P_n : \varepsilon(x) = 1\} \le s_p$ and $0 \le \delta_p(n) \le \gamma_p(n)$. Hence $N_p(n) - Q_p^2 p\, r_p(n) = 2Q_p\gamma_p(n) + \delta_p(n) \le (2Q_p + 1)s_p$. Since $Q_p s_p < Q_p d_p \le L+1$ and $s_p < d_p \le L+1$,

$$(2Q_p + 1)s_p = 2Q_p s_p + s_p < 2(L+1) + (L+1) = 3(L+1).$$

Summing over the (at most $\pi(L+1)$) primes with $d_p \le L+1$ gives the bound; the prime number theorem gives $\pi(L+1) \sim (L+1)/\log(L+1)$. The ≤ 2 known Wieferich primes contribute, separately and trivially, $O(L)$ each. ☐

Remark 4.3 (Numerical confirmation). Script `verify_residual.py` computes $R1(n)$ exactly for the worst n at every dyadic scale up to $n \approx 2^{22}$ ($L = 13, \dots, 21$). The maximal prime ever carrying a boundary term in this range is 7 (far below the bound $L+1$), and the measured $R1 \in \{0, 4, 8, \dots, 32\}$ stays an order of magnitude below the proved bound $3(L+1)\pi(L+1) \in [234, 528]$ over the same range.

Combining Theorem 3.4 and Proposition 4.2: for every odd n, the total contribution of primes with $d_p \le L+1$ (*all* primes other than the ≤ 2 known Wieferich exceptions) is at most

$$0.3205\,(L+1)^2 + O\!\left(\frac{L^2}{\log L}\right) + O(L) \;=\; \big(0.3205 + o(1)\big)(L+1)^2,$$

uniformly in n – an elementary, rigorous, complete closure leaving a margin of $\approx 0.18\,T$ before the threshold $T = (L+1)^2$ of Proposition 2.2.

5 Lemma M: deterministic dissolution of the worst case

What remains is the contribution of primes with $d_p > L+1$ (no complete period inside $\{0,\dots,L\}$); by Proposition 4.1 (contrapositive) this is exactly the set of primes $p > L+1$ that are non-Wieferich, together with the (negligible) Wieferich exceptions. Write

$$R2(n) := \sum_{\substack{p\le\sqrt{n}\\ d_p>L+1}} N_p(n).$$

By the trivial bound $N_p(n) \le L+1$ (distinct $2^l \bmod p^2$, no period), combined with the prime-counting bound $\#\{p \le \sqrt{n}\} \sim \sqrt{n}/\log\sqrt{n}$, elementary counting alone gives only $R2(n) \le (L+1)\sqrt{n}$, hopelessly larger than $T = (L+1)^2$. The empirical data ([11]) shows $R2(n)$ is in fact always $\ll L^2$ and concentrated on a bounded range of primes just above L; this section isolates, deterministically and without reference to any particular n, exactly how much of this is provable.

We restrict attention to the range $p \in (L, L^2]$, which the data identifies as carrying the dominant share of $R2$; the remaining two sub-ranges of $R2$ are discussed in §7.

Definition 5.1. For a prime p, set $M_p := \max_n N_p(n)$ (n ranging over odd integers, with L implicitly determined by n – equivalently, M_p is the maximal number of representations of any residue $n \bmod p^2$ as $2^l + 2^m$ with $0 \le l, m \le L$, for the relevant L). This is an n-independent quantity.

Lemma 5.2 (Lemma M – deterministic dissolution)**.** *For* every *odd* n*,*

$$\sum_{p\in(L,L^2]} N_p(n) \;\le\; \sum_{p\in(L,L^2]} M_p.$$

Proof. $N_p(n) \le \max_{n'} N_p(n') = M_p$ for every p, by definition; sum over p. □

This trivial-looking statement is the key structural gain of this section: *it eliminates the worst-case-n problem entirely* for this range. If $\sum_{p\in(L,L^2]} M_p = o(L^2)$, this range contributes $o(L^2)$ to *every* n simultaneously, deterministically – no averaging, no exceptional set.

5.1 The unconditional half: the Sidon decomposition

If $2^i + 2^j \equiv 2^{i'} + 2^{j'} \pmod{p^2}$ has no solution with $\{i,j\} \ne \{i',j'\}$ (both pairs ranging over $\{0,\dots,L\}$, L ranging implicitly over all values relevant to M_p), we call p *Sidon*; then every value $n \bmod p^2$ has at most one unordered representation, so the ordered multiplicity is at most 2 (the pair and its swap).

Proposition 5.3 (Unconditional Sidon term)**.**

$$\sum_{p\in(L,L^2]} M_p \;=\; \underbrace{2\big(\pi(L^2)-\pi(L)\big)}_{\textit{Sidon contribution}} \;+\; \underbrace{\sum_{p\in(L,L^2]\text{ non-Sidon}} (M_p-2)}_{\textit{excess}},$$

and the Sidon contribution alone satisfies $2(\pi(L^2)-\pi(L)) = O(L^2/\log L) = o(L^2)$, unconditionally *(immediate from the prime number theorem; no structural input about* $\{2^l \bmod p^2\}$ *is needed for this half).*

Proof. For Sidon p, $M_p = 2$ exactly (a representation and its swap; $M_p \geq 2$ trivially once $L \geq 1$). The displayed identity is then a rearrangement of the sum. The bound on the Sidon term is the prime-counting estimate $\pi(L^2) - \pi(L) \sim L^2/(2\log L)$. $\square$

The remaining *excess* term – the total multiplicity overshoot summed over non-Sidon primes in $(L, L^2]$ – is exactly the open quantity discussed in §7 (named M″ there). Brute-force counting of it already fails at the first step: the crude bound $M_p - 2 \leq L - 1$ (trivial, since $M_p \leq L+1$) combined with the trivial bound on the number of non-Sidon primes ($\leq \pi(L^2) - \pi(L)$) gives only $\Theta(L^2)$, not $o(L^2)$ – the excess is genuinely smaller than this in the data (by roughly a factor $1/\log L$), but proving it requires controlling the *distribution* of the multiplicities M_p, not just their existence.

5.2 The order sum is unconditionally $o(L^2)$

We isolate one half of the excess that *is* unconditional. Split the range by the multiplicative order of 2: call $p \in (L, L^2]$ *Type A* if $e_p \leq L$ and *Type B* if $e_p > L$.

Proposition 5.4 (Order sum)**.** $\displaystyle\sum_{p\in(L,L^2]} \left\lfloor \frac{L}{e_p} \right\rfloor = O\Big(\frac{L^2}{\log L}\Big) = o(L^2).$

Proof. $\lfloor L/d \rfloor = \#\{j \in [1, L] : d \mid j\}$, and $e_p \mid j \iff p \mid 2^j - 1$, so $\sum_p \lfloor L/e_p \rfloor = \sum_{j\leq L} \#\{p \in (L, L^2] : p \mid 2^j - 1\}$. The primes $p > L$ dividing $2^j - 1$ have product $\leq 2^j - 1 < 2^j$; if there are m of them then $L^m < 2^j$, so $m < j \log 2/\log L$. Summing, $\sum_{j\leq L} j \log 2/\log L = (\log 2/\log L) \cdot L(L + 1)/2 = O(L^2/\log L)$. $\square$

The gain over the trivial bound (which counts each $2^j - 1$ with its full $O(j)$ prime factors and gives $\Theta(L^2)$) is a factor $\log L$, and it comes *entirely* from the constraint $p > L$: large primes cannot divide $2^j - 1$ too often. This is the only place the lower endpoint of the range is used.

5.3 Classification of $(L, L^2]$: the involution structure

Each pair contributes $2^l + 2^m = 2^l(1 + 2^{m-l})$; a fibre's arithmetic is governed by the *gap values* $1 + 2^\delta$ ($\delta = |m - l|$), which satisfy an exact identity.

Lemma 5.5 (Involution)**.** *For every δ,* $1 + 2^{\,e_p-\delta} \equiv 2^{-\delta}(1 + 2^\delta) \pmod p$.

Proof. $2^{-\delta}(1 + 2^\delta) = 2^{-\delta} + 1 = 2^{\,e_p-\delta} + 1$, using $2^{e_p} \equiv 1 \pmod p$. $\square$

Thus δ and $e_p - \delta$ send $1 + 2^\delta$ into the same coset of $\langle 2\rangle$ mod p: the gap values pair up under $\delta \mapsto e_p - \delta$. Using the p-adic refinement $2^{e_p} \equiv 1 + w_p p \pmod{p^2}$ (with $w_p \neq 0$ exactly for non-Wieferich p), the representations of a fixed residue n mod p^2 organise into arithmetic progressions in δ of common difference a multiple of e_p, giving a per-class bound.

Proposition 5.6 (Per-class bound)**.** *Let $p \in (L, L^2]$ be non-Wieferich. For any residue n mod p^2, the pairs $(l, m) \in \{0, \ldots, L\}^2$ with $2^l + 2^m \equiv n$, grouped by δ mod e_p, form at most $\lfloor L/e_p \rfloor + 1$ pairs per class. Hence*

$$M_p \;\leq\; \kappa_p\,(\lfloor L/e_p \rfloor + 1), \qquad \kappa_p := \#\{\textit{active classes } \delta \bmod e_p \textit{ in a maximal fibre}\}.$$

Proof sketch. Since $d_p = p\,e_p > L$, for each l there is at most one $m \in \{0, \ldots, L\}$ with $2^m \equiv n - 2^l \pmod{p^2}$. Within a class ($l$ mod e_p, δ mod e_p), writing $l = l_0 + e_p a$, $\delta = \delta_0 + e_p b$, the congruence reduces (via $2^{e_p} \equiv 1 + w_p p$, $w_p \neq 0$) to a single linear congruence $aX + bY \equiv Z \pmod p$ in the triangle $\{a, b \geq 0,\ a + b \leq (L - l_0 - \delta_0)/e_p\}$ of side $T \leq L/e_p < p$; as $T < p$, each b admits at most one a, giving $\leq \lfloor L/e_p \rfloor + 1$ pairs. Sum over the κ_p active classes. (Verified with no exceptions for all $p \in (L, L^2]$, $L \leq 160$; non-degeneracy of the linear form is exactly the non-Wieferich condition.) $\square$

Empirically $\kappa_p \leq 2$ for *genuinely small* order ($e_p \ll L$: the active classes form at most one involution orbit plus a fixed point), whereas for *boundary* primes $e_p \approx L$ (where 2 is near-primitive) κ_p behaves like a Poisson maximum and grows slowly (measured ≤ 6 for $L \leq 200$). We do not prove a bound; we isolate as hypothesis (TA) the mild growth condition $\max_{e_p \leq L} \kappa_p = o(\log L)$. Since $\sum_p \lfloor L/e_p \rfloor$ already carries a $1/\log L$ saving (Proposition 5.4), this suffices:

$$\sum_{p \in (L, L^2]:\, e_p \leq L} M_p \ \leq\ \Big(\max_{e_p \leq L} \kappa_p\Big)\Big(\sum_p \lfloor L/e_p \rfloor + \pi(L^2)\Big) \ =\ o(\log L) \cdot O\Big(\tfrac{L^2}{\log L}\Big) \ =\ o(L^2) \quad \text{[under (TA)]}.$$

A partial unconditional bound on κ_p. The empirical $\kappa_p \leq 2$ is not needed in full: a theorem of García–Voloch on additive representations in a multiplicative subgroup gives a non-trivial bound directly.

Proposition 5.7 (Unconditional $\kappa_p \ll e_p^{2/3}$, via García–Voloch)**.** *For $G = \langle 2 \rangle \bmod p$ with $|G| = e_p < (p-1)/((p-1)^{1/4} + 1)$ (so $e_p \lesssim p^{3/4}$) and any $b \not\equiv 0 \pmod p$, one has $r_p(b) := \#\{(u, v) \in G^2 : u + v \equiv b\} \leq 4\, e_p^{2/3}$ (García–Voloch [7], in the exact form of Konyagin's Theorem 2.1 [8]; Heath-Brown gave a Stepanov-method proof, and Heath-Brown–Konyagin the energy $T_2(G) \ll e_p^{5/2}$ for $e_p \leq p^{2/3}$). Reducing a maximal fibre modulo p sends its κ_p active classes to distinct solutions of $u + v \equiv n \bmod p$ with $u, v \in G$ (two pairs in different classes $\delta \bmod e_p$ give different gap values $1 + 2^\delta$, hence different representations mod p); so $\kappa_p \leq r_p(n \bmod p) \ll e_p^{2/3}$ unconditionally, whence*

$$\sum_{p \in (L, L^2]:\, e_p \leq L} M_p \ =\ O\left(\frac{L^{8/3}}{\log L}\right).$$

Proof. With $M_p \leq \kappa_p(\lfloor L/e_p \rfloor + 1)$ and $\#\{p > L : e_p = e\} \ll e/\log L$ (the counting behind Proposition 2.4), $\sum M_p \ll (\log L)^{-1} \sum_{e \leq L} e^{2/3}(L + e) \asymp L^{8/3}/\log L$. Two ranges use the trivial $M_p \leq L + 1$ instead. The size hypothesis fails only for $e_p > p^{3/4}$, i.e. $p < e_p^{4/3} \leq L^{4/3}$: a band of $\leq \pi(L^{4/3}) = O(L^{4/3}/\log L)$ primes, total $O(L^{7/3}/\log L)$. The excluded residues $b \equiv 0 \pmod p$ occur only when $p \mid n$, for $\leq \log n/\log L = O(L/\log L)$ primes, total $O(L^2/\log L)$; the p^2-degenerate residue ($2^l + 2^m \equiv 0 \bmod p^2$) is unreachable in range for Type A, as it needs $m - l \asymp \mathrm{ord}_{p^2}(2)/2 > L$. Both bands are $o(L^{8/3}/\log L)$, absorbed. (Numerically the band holds $8, 15, 16, 23, 43$ primes for $L = 60, \ldots, 300$, always $\max p < L^{4/3}$, and the ratio $r_p(b)/e_p^{2/3} < 1$ with no drift to $e_p \approx 900$; scripts `typeA_garcia_voloch_check.py`, `typeA_exceptional_band.py`.) $\square$

This is short of the target $o(L^2)$ by a factor $L^{2/3}$. Hypothesis (TA) is thus now the sharper open question of whether the exponent $2/3$ can be lowered to $o(1)$ on the special fibres $1 + 2^\delta$ – a rigidity that the generic Stepanov argument ignores but the involution orbit structure may supply.

5.4 The Type B residual as a factorial-moment condition

For Type B primes ($e_p > L$) the fibres carry no progression ($\lfloor L/e_p \rfloor = 0$) and M_p is governed by coincidences alone. The sharp reduction is through factorial moments. For $k \geq 2$ put

$$E_k^{\mathrm{tot}} := \sum_{\substack{p \in (L, L^2] \\ e_p > L}} \sum_r \binom{N_p(r)}{k}, \qquad E_k^{\mathrm{null}} := \sum_{\substack{p \in (L, L^2] \\ e_p > L}} \frac{\binom{P}{k}}{(p^2)^{k-1}}, \quad P := \tbinom{L+2}{2},$$

E_k^{null} being the k-th factorial moment of the multinomial (P balls in p^2 bins) summed over the range – the "random" benchmark.

Lemma 5.8 (Factorial-moment tail). *$\#\{p \in (L, L^2] : e_p > L,\ M_p \geq k\} \leq E_k^{\mathrm{tot}}$, and $E_k^{\mathrm{null}} \sim L^3/\big(2^k k!\,(2k-3)\log L\big)$, which crosses 1 at $k^* \sim 3\log L/\log\log L$.*

Proof. Each r with $N_p(r) \geq k$ contributes $\binom{N_p(r)}{k} \geq 1$, and each such p has at least one such r; combine. The asymptotic uses $\binom{P}{k} \sim P^k/k!$, $P \sim L^2/2$, and $\sum_{p>L} p^{-(2k-2)} \sim L^{-(2k-3)}/((2k-3)\log L)$. □

Theorem 5.9 (Conditional every-n variant). *Suppose:*

(TB) *there is an absolute C with $E_k^{\mathrm{tot}} \leq C^k E_k^{\mathrm{null}}$ for all $k \geq 2$; only the tail $k > k^* = \lfloor 3\log L/\log\log L \rfloor$ is actually used, the range $2 \leq k \leq k^*$ being unconditional (see the proof);*

(TA) $\displaystyle\max_{p\in(L,L^2]:\, e_p \leq L} \kappa_p = o(\log L)$;

(WB) $\sum_{p\in(L^2,\sqrt{n/2}]} N_p(n) = o(L^2)$ *uniformly in n.*

Then the two-powers variant holds for every odd $n > n_0$.

Proof. By Proposition 2.2 it suffices that $\sum_{3\leq p\leq\sqrt{n}} N_p(n) < T$. The range $p \leq L+1$ contributes $(0.3205 + o(1))T$ (Theorem 3.4, Proposition 4.2); $(\sqrt{n/2}, \sqrt{n}]$ contributes $O(L)$ (Kalinin [2]); $(L^2, \sqrt{n/2}]$ contributes $o(L^2)$ by (WB). For $(L, L^2]$, by Lemma 5.2 it suffices that $\sum_{p\in(L,L^2]} M_p = o(L^2)$; Type A is $o(L^2)$ under (TA) by §5.3, and Type B is, under (TB) and Lemma 5.8,

$$\sum_{e_p>L}(M_p-1) = \sum_{k\geq 2}\#\{M_p \geq k\} \;\leq\; \underbrace{k^*\,\pi(L^2)}_{\text{unconditional}} \;+\; \underbrace{\sum_{k>k^*} C^k E_k^{\mathrm{null}}}_{\text{needs (TB), } k>k^*} \;=\; O\Big(\tfrac{L^2}{\log\log L}\Big) = o(L^2).$$

The first term is unconditional: each $\#\{M_p \geq k\} \leq \pi(L^2)$, and there are $\leq k^* = O(\log L/\log\log L)$ of them, giving $O(L^2/\log\log L)$. Only the tail $k > k^*$ invokes (TB), and there $\sum_{k>k^*} E_k^{\mathrm{null}} = o(1)$ since E_k^{null} has already crossed 1 at k^*. Summing the ranges gives $(0.3205 + o(1))T + o(L^2) < T$ for n large. □

Hypothesis (TB) is the single genuinely analytic input, and it is equivalent to a *square-sieve / large-sieve* bound. Since $\{2^k\}$ is a Sidon set over $\mathbb{Z}$, E_k^{tot} counts k-fold square-divisor incidences $\#\{p > L : p^2 \mid \sum_i \varepsilon_i 2^{j_i}\}$ of power combinations; the natural tool is Heath-Brown's square sieve [6]. Its main term reproduces E_k^{null} (the correct order, obtained by averaging over p: per-prime bounds are a factor L too weak), and its error term is a *mean-square* bound on the incomplete character sums $\sum_{j\leq L}\chi(2^j)$ over $\{2^j \bmod p^2\}$. These are lacunary, so on the torus $\|\sum a_j e(2^j\theta)\|_{2k} \ll \sqrt{k}\,\|a\|_2$ (Rudin [10]); the corresponding $\bmod\, p^2$ bound holds empirically but reaches size $\sim L$ (not $O(1)$) on quasi-trivial characters, so a rigorous proof needs the full large sieve, not a pointwise estimate. This is the same incomplete-exponential-sum core as the fourth toolkit of §7, now in mean-square rather than sup-norm form. Empirically (TB) holds with margin:

L	$E_2^{\mathrm{tot}}/E_2^{\mathrm{null}}$	$E_3^{\mathrm{tot}}/E_3^{\mathrm{null}}$	$E_4^{\mathrm{tot}}/E_4^{\mathrm{null}}$
80	0.91	0.91	0.40
120	0.95	0.70	0.35
160	0.95	0.86	0.67

All ratios are ≤ 1: the arithmetic fibres are *sub*-multinomial at every order, and E_k^{tot} already vanishes by $k = 5$ at these L (well below the multinomial prediction). Restricting to Type B removes the super-concentrated Type A fibres, whose energy is order-driven, not random. Note that no independence or negative-association hypothesis is used: the passage from the moments to the maximum is the elementary first-moment (Markov) bound of Lemma 5.8.

Remark 5.10 (Which moment is binding). It is worth stating precisely which part of (TB) is used, to avoid a common misreading. The Type B contribution is $\sum_{e_p>L}(M_p - 1) = \sum_{k\geq 2} \#\{p : M_p \geq k\}$, and *every* term is bounded trivially by $\#\{p : M_p \geq k\} \leq \pi(L^2) = o(L^2)$; in particular the $k = 2$ term is free, and the individual energy $E_2^{\mathrm{tot}} = \sum_p \Delta_p$ is *not* $o(L^2)$ (it is $\Theta(L^3/\log L)$, and need not be small). What must be shown is only that the sum has $O(\pi(L^2))$ nonzero terms, i.e. that the *maximum* multiplicity satisfies $\max_{e_p>L} M_p = o(\log L)$; then $M_B'' \leq (\max M_p)\,\pi(L^2) = o(L^2)$. Thus the binding instance of (TB) is the *large-k* one, $k \asymp \log L/\log\log L$ (a large-deviation statement about the maximum), not the variance $k = 2$. The $k = 2$ case is merely the cleanest window onto the mechanism forcing sub-Poisson behaviour, namely that $\{2^k\}$ is Sidon over $\mathbb{Z}$, so a collision mod p^2 forces $p^2 \mid N$ with $N \neq 0$, an event of density $\sum_p 1/p^2$ (verified: the density heuristic reproduces E_2^{tot} to within 5%, stably in L). This mechanism is base-independent: the same holds for $\{g^k\}$, any $g \geq 2$.

5.5 The additive sup-norm, and why (TB) is closed by circularity

The square-sieve error term above is phrased through *multiplicative* characters χ of $(\mathbb{Z}/p^2)^\times$, and there $\sum_{j\leq L} \chi(2^j)$ is a geometric progression that reaches size $\sim L$ when χ is quasi-trivial on $\langle 2\rangle$ – no cancellation. The *additive* side behaves better, and unconditionally so. Write $S(\xi) := \sum_{j\leq L} e_{p^2}(\xi 2^j)$.

Theorem 5.11 (Additive sup-norm, Bourgain–Chang [9], Cor. 4.5)**.** *Let $q = \prod p_\alpha^{\nu_\alpha}$ have a bounded number of prime factors and $\theta \in \mathbb{Z}_q^\times$ with $\mathrm{ord}_p(\theta) > q^\delta$ for every $p \mid q$. Then for $t > q^\delta$, $\max_{\xi\in\mathbb{Z}_q^\times} |\sum_{s\leq t} e_q(\xi\theta^s)| < t^{1-\varepsilon}$ with $\varepsilon = \varepsilon(\delta) > 0$ absolute. For $q = p^2$, $\theta = 2$, $t = L + 1$ and $p \in (L, L^2]$ Type B, take $\delta = \frac{1}{5}$: then $e_p > L \geq q^{1/5}$ and $L + 1 > q^{1/5}$, so*

$$|S(\xi)| < (L + 1)^{1-\varepsilon} \quad (p \nmid \xi), \qquad |S(p\eta)| < (L + 1)^{1-\varepsilon'} \ (\eta \not\equiv 0),$$

the latter by reduction $\bmod p$ (order $e_p > \sqrt{p}$).

Numerically $\max_{p\nmid\xi} |S(\xi)|/(L + 1) = 0.37, 0.31, 0.24$ at $L = 40, 80, 120$ on Type B primes – well below 1 and decreasing (script `typeB_supnorm_check.py`). The counter-example (4.23) in [9], $\theta = 1 + p$ (order 1 mod p, hence no cancellation), is exactly why Type A ($e_p \leq L$, small order $\bmod p$) is *not* covered by Theorem 5.11: the Type A/Type B split is forced, Type A being the García–Voloch regime (Proposition 5.7) and Type B the Bourgain–Chang one.

One might hope Theorem 5.11 closes (TB). It does not, for a structural reason worth recording.

Proposition 5.12 (The sup-norm cannot close the conversion)**.** *Expanding E_k^{tot} by additive orthogonality gives, per prime, $\sum_r N_p(r)^k = (p^2)^{1-k} \sum_{\xi_1+\cdots+\xi_k\equiv 0} \prod_i S(\xi_i)^2$. Bounding every nonzero frequency by $|S| < B := (L+1)^{1-\varepsilon_0}$ leaves each extra nonzero frequency a factor $B^2 = (L+1)^{2-2\varepsilon_0} > 1$ (for any $\varepsilon_0 < 1$): the all-nonzero term is $\gtrsim (L + 1)^{2k-2-2(k-2)\varepsilon_0} L^2/(2^k k! \log L)$, which at $k \sim k^*$ is $\exp\bigl(c(1 - \varepsilon_0)\log^2 L/\log\log L\bigr) \gg L^2$. Sup-norm alone is insufficient regardless of ε_0.*

Proposition 5.13 (Circularity: the missing bound *is* (TB))**.** $\sum_{\xi \bmod p^2} |S(\xi)|^{2k} = p^2 E_k^{\mathrm{add}}$*, where $E_k^{\mathrm{add}} = \#\{(j_i, j_i')_{i\leq k} : \sum_i 2^{j_i} \equiv \sum_i 2^{j_i'} \bmod p^2\}$ is the additive energy of $\{2^j\}$ – exactly the quantity*

E_k^{tot} *measures. Thus the moment (L^{2k}) bound that would close the conversion of Proposition 5.12 is logically equivalent to (TB) itself. The sup-norm (L^∞) is available and unconditional; the missing L^{2k} bound is the conjecture. Hence (TB) is a genuine open estimate, not a statement awaiting the right reference.*

6 An unconditional theorem for almost all n

This section proves a genuine, unconditional theorem about the same range $p \in (L, L^2]$, by trading the (currently unavailable) worst-case control of §5 for an average-case control, via a second moment and Markov's inequality. We state its scope precisely: it controls $p \in (L, L^2]$ only: see the discussion at the end of this section and §7 for what this does, and does not, give for the full two-powers variant.

Proposition 6.1 (Half for free)**.** *For every odd n, $\#\{p \in (L, L^2] : N_p(n) \ge 1\} \le \pi(L^2) = o(L^2)$, unconditionally (there are simply not enough primes in $(L, L^2]$ for the mere* count *of contributing primes to reach L^2).*

Definition 6.2. $C(n) := \sum_{p\in(L,L^2]} N_p(n)\big(N_p(n) - 1\big)$ – the *coincidence term*: twice the number of triples (p, P, P') with $P \ne P'$ two pairs in $\{0, \dots, L\}^2$ both summing to n mod p^2.

By Proposition 6.1, $\sum_{p\in(L,L^2]} N_p(n) \le \pi(L^2) + C(n)$ for every n, so it remains to control $C(n)$.

Proposition 6.3 (Second moment)**.** *Let $\overline{C}$ denote the average of $C(n)$ over residues n modulo $\prod_{p\in(L,L^2]} p^2$ (equivalently, over a full period in each modulus separately, by CRT). Then*

$$\overline{C} = \sum_{p\in(L,L^2]} \frac{\Delta_p}{p^2}, \qquad \Delta_p := \#\{(P, P') : P \ne P',\ \mathrm{sum}(P) \equiv \mathrm{sum}(P') \pmod{p^2}\}.$$

For $p \in (L, L^2]$, $\Delta_p \approx \binom{L+2}{2}^2/p^2$ for a quasi-random set of $\binom{L+2}{2}$ sums in $\mathbb{Z}/p^2$ (confirmed numerically, no exceptional clustering observed), giving

$$\overline{C} \;\approx\; \frac{(L+2)^4}{4} \sum_{p>L} \frac{1}{p^4} \;\approx\; \frac{L^4}{4} \cdot \frac{1}{3L^3 \log L} \;=\; \frac{L}{12 \log L} \;=\; o(L^2).$$

Remark 6.4. The approximation $\Delta_p \approx (\#\text{pairs})^2/p^2$ is the quasi-random heuristic for collision counts and is *not* proved here in full rigor for every p; it is confirmed numerically ($\overline{C} \approx 0.44, 0.43, 0.51, 0.96, 0.52$ for $L = 16, 20, 24, 28, 32$, matching the predicted $O(L/\log L)$ growth closely). What *is* fully rigorous, without this heuristic, is the cruder bound obtained by bounding Δ_p trivially by $\binom{L+2}{2}^2$ and summing $1/p^2$ over $p > L$ via Mertens: this already gives $\overline{C} = O(L^2 \cdot 1/L) = O(L)$, which still suffices for what follows (any bound $\overline{C} = o(L^2)$ does).

Theorem 6.5 (Almost all n, on the range $(L, L^2]$)**.** *For every $\eta > 0$, the density of odd $n \le 2^{L+1}$ for which $\sum_{p\in(L,L^2]} N_p(n) \ge \pi(L^2) + \eta L^2$ is at most $\overline{C}/(\eta L^2) = O(1/(L \log L)) \to 0$. In particular, taking $\eta = 0.4$: for all but a density-$O(1/(L \log L))$ exceptional set of odd n,*

$$\sum_{p\in(L,L^2]} N_p(n) \;<\; \pi(L^2) + 0.4\, L^2 \;=\; (0.4 + o(1))(L+1)^2.$$

Proof. Markov's inequality applied to $C(n) \geq 0$. It remains to justify that the mean of $C(n)$ over odd $n \leq 2^{L+1}$ equals $\overline{C}$ (defined by CRT over the astronomically larger modulus $\prod_{p\in(L,L^2]} p^2$) up to $o(L^2)$. Each condition $p^2 \mid n - 2^l - 2^m$ meets the interval $[2^L, 2^{L+1})$ in a fixed residue class, with frequency $\frac{1}{p^2} + O(2^{-L})$; summing the $O(2^{-L})$ errors over the $\leq \binom{L+2}{2}$ pairs and the $\leq \pi(L^2)$ primes gives a total discrepancy $\ll L^2 \cdot \pi(L^2) \cdot 2^{-L} \ll L^6\, 2^{-L} = o(1)$ between the interval mean of $C(n)$ and $\overline{C}$. (Restricting to odd n only halves the interval and changes nothing, since $2^l + 2^m$ is even and p^2 is odd.) Hence the interval mean is $\overline{C} + o(1) = o(L^2)$, and Markov applies. □

6.1 What this gives, and what it does not, for the full variant

Combining Theorem 3.4, Proposition 4.2 (closing $p \leq L+1$, unconditionally, for *every* n, contributing $(0.3205 + o(1))T$) with Theorem 6.5 (closing $p \in (L, L^2]$, for *almost all* n, contributing $(0.4 + o(1))T$) gives, for that same density-$1 - O(1/(L\log L))$ set of n:

$$\sum_{\substack{p\leq\sqrt{n}\\ p\leq L+1 \text{ or } p\in(L,L^2]}} N_p(n) \;<\; (0.7205 + o(1))\, T,$$

leaving an explicit margin of $\approx 0.28\,T$ before the threshold T of Proposition 2.2. It remains to control the range $p \in (L^2, \sqrt{n/2}]$ (wall (B) of §7); for that range, on average, a *first* moment already suffices, as we now show – so the almost-all-n statement is in fact complete.

6.2 Wall (B) on average: a first-moment bound

Theorem 6.6 (Wall (B) on average: first moment at a raised threshold)**.** *For $N = 2^L$, the mean over $n \in [N, 2N)$ of $B(n) := \sum_{p\in(L^2,\sqrt{n/2}]} N_p(n)$ is $O(1/\log L)$. Consequently, by Markov, for every $\eta > 0$,*

$$\#\{n \in [N, 2N) :\; B(n) \geq \eta L^2\} \;\leq\; N \cdot O\Big(\frac{1}{\eta L^2 \log L}\Big) + o(N).$$

Two useful thresholds: at $\eta L^2 \to 1$ (i.e. threshold 1), all but a density-$O(1/\log L)$ set of n have no *pair (l, m) with $n - 2^l - 2^m$ divisible by a square p^2, $p \in (L^2, \sqrt{n/2}]$; at any fixed $\eta > 0$, all but a density-$O(1/(L^2 \log L))$ set have $B(n) < \eta L^2$. The latter is what the $0.28\,T$ margin permits, and it is a factor L^2 smaller than the former.*

Proof. The mean is bounded by summing over pairs and primes:

$$\sum_{n\in[N,2N)} B(n) \leq \sum_{(l,m)} \sum_{p>L^2} \#\{n \in [N, 2N) : n \equiv 2^l{+}2^m \pmod{p^2}\} \leq (L{+}1)^2\Big(N \sum_{p>L^2} \tfrac{1}{p^2} {+} \pi(\sqrt{N/2})\Big),$$

since each residue class modulo p^2 meets $[N, 2N)$ in at most $N/p^2 + 1$ points and the relevant primes are $\leq \sqrt{N/2}$. By Mertens, $\sum_{p>L^2} 1/p^2 = O(1/(L^2 \log L))$, so the first term is $(L + 1)^2 \cdot N \cdot O(1/(L^2 \log L)) = N \cdot O(1/\log L)$; the second is $(L + 1)^2 \pi(\sqrt{N/2}) = O(L^2 \sqrt{N}/\log N) = o(N)$ since $N \gg L^4$. Dividing by N gives mean $O(1/\log L)$; Markov's inequality at level ηL^2 gives the displayed density. (The threshold-1 form is the special case, recovering the union bound.) □

Remark 6.7 (Higher moments: a genuine but conditional improvement)*.* Over a full CRT period $\prod_p p^2$ the counters $N_p(n)$ for distinct p are exactly independent, so the moments of $B(n)$ and of the range-$(L, L^2]$ collision count $C(n) = \sum_p N_p(N_p - 1)$ factor over p, and the interval bridge stays valid for the $2k$-th moment as long as the modulus $(L^2)^{2k} \ll 2^L$, i.e. $k \ll L/\log L$. This suggests exceptional density $O(L^{-A})$ for every fixed A. We caution, however, that this is *not* unconditional: the variance is $\mathrm{Var}(C) = \sum_p \mathrm{Var}(N_p(N_p-1)) \le M_{\max}^2\,\overline{C}$ where $M_{\max} = \max_p M_p$, and with only the trivial $M_{\max} \le L+1$ this gives $\mathrm{Var}(C) = O(L^2\overline{C})$ and Chebyshev *no* gain over Markov; a genuine gain (numerically $\mathrm{Var}(C)/\overline{C} \approx 2.1$–$2.9$ to $L = 36$, giving density $O(1/(L^3 \log L))$) requires $M_{\max} = o(L)$, or more precisely an unconditional bound on the fourth-moment sum $\sum_p p^{-2} \sum_r [c_r(c_r-1)]^2$ – a higher square-sieve estimate of the same nature as (TB). So the moments beyond the first are controlled by the same open input as (M″) itself; only the first moment (Markov) is unconditional. The wide range (B), where $N_p \in \{0,1\}$ generically, is closer to Bernoulli and its second moment is more plausibly unconditional, but we do not pursue it. The first moment already suffices for the corollary.

This is the structural counterpart of the impossibility noted in §7: there is simply not enough mass in $\sum_{p>L^2} 1/p^2$ (which is $< \sum_{p\ge 3} 1/p^2 = 0.2022\ldots$) for the wide range (B) to matter on average, even though it is the numerically dominant range. Numerically (script `typeB_wall_first_moment.py`), the fraction of n with a nonzero (B)-contribution is $0\%, 3.4\%, 5.4\%$ at $L = 15, 18, 22$, average contribution ≤ 0.056, in line with $(L+1)^2 \sum_{p>L^2} 1/p^2 \approx 0.13$ (slowly decreasing).

6.3 The perfect-square range, for almost all n

For the every-n statement the range $(\sqrt{n/2}, \sqrt{n}]$ is closed by Kalinin [2] (§7). For the *almost-all* statement it needs no such input: a first moment suffices, keeping the unconditional theorem entirely self-contained.

Proposition 6.8 (Perfect-square range, almost all n)**.** *For all but $O\big((L+1)^2\, 2^{-L/2}\big) = o(2^L)$ odd $n \in [2^L, 2^{L+1})$, no pair (l, m) has $n - 2^l - 2^m$ a perfect square; a fortiori the range $(\sqrt{n/2}, \sqrt{n}]$ contributes* 0*.*

Proof. For $p > \sqrt{n/2}$ and $p^2 \mid k$ with $0 < k < n \le 2p^2$, the only multiple of p^2 available is $k = p^2$; so any contribution from this range requires $n - 2^l - 2^m$ to be a perfect square. Such an n equals $2^l + 2^m + s^2$ for some $l, m \le L$ and $s \ge 0$. The number of triples (l, m, s) with $2^l + 2^m + s^2 \in [2^L, 2^{L+1})$ is $\sum_{l,m} \#\{s : s^2 \in [2^L - 2^l - 2^m,\ 2^{L+1} - 2^l - 2^m)\} \le (L+1)^2(2^{L/2}+1)$, since *any* interval of length X holds at most $\sqrt{X}+1$ squares (consecutive squares in $[a, a+X)$ satisfy $\sqrt{a+X} - \sqrt{a} \le \sqrt{X}$); here $X = 2^L$, and the interval may start near 0 when $2^l + 2^m \approx 2^L$, so the universal bound is needed. Hence the exceptional set has size $\le (L+1)^2(2^{L/2}+1)$, of density $O((L+1)^2\, 2^{-L/2}) \to 0$. □

6.4 The complete almost-all-n theorem

Corollary 6.9 (Two-powers variant for almost all n, unconditional)**.** *For all but a density-$O(1/(L \log L))$ set of odd $n \le 2^{L+1}$, the two-powers variant holds.*

Proof. Outside the union of the exceptional sets – never (for $p \le L+1$, by Theorem 3.4 and Proposition 4.2, deterministically $\le (0.3205 + o(1))T$); density $O(1/(L \log L))$ (for $(L, L^2]$, by Theorem 6.5 at $\eta = 0.4$, contributing $< (0.4 + o(1))T$); density $O(1/(L^2 \log L))$ (for $(L^2, \sqrt{n/2}]$, by Theorem 6.6 at $\eta = 0.1$, contributing $< 0.1\,T$); density $O((L+1)^2\, 2^{-L/2})$ (for $(\sqrt{n/2}, \sqrt{n}]$, contributing 0, by Proposition 6.8) – we get $\sum_{p\le\sqrt{n}} N_p(n) < (0.8205 + o(1))T < T$, whence a squarefree pair exists by Proposition 2.2. The union of exceptional sets has density $O(1/(L \log L)) \to 0$, dominated by $(L, L^2]$; in particular the unconditional theorem now uses no external input for any range. □

Thus the only gaps to a *full* (every-n) proof are the worst-case statements (M″) and (B) of §7; the almost-all-n statement is unconditional and complete. Raising the wall-(B) threshold from 1 to

ηL^2 (which the $0.28\,T$ margin permits) has moved the binding term in the exceptional density from (B) to the range $(L, L^2]$, at $O(1/(L\log L))$ – a factor-L improvement over the naive $O(1/\log L)$. A further conditional improvement to $O(L^{-A})$ (any A) is available via higher moments, but rests on a fourth-moment bound of the same difficulty as (M″) (see the preceding remark); the unconditional density remains $O(1/(L\log L))$.

7 What remains: the open links

We now state, as precisely as possible, the two gaps left open by the above, together with where each brute-force argument fails and by how much.

7.1 The range $(\sqrt{n/2}, \sqrt{n}]$: closed, by Kalinin (2026)

If $p \in (\sqrt{n/2}, \sqrt{n}]$ and $N_p(n) \geq 1$ with the further coincidence $2^l = 2^m$ (i.e. a perfect-square term $n - 2^{l+1}$), Kalinin [2] proves (his Theorem 4, building on a multiplicity bound $|\{k \geq 1 : n - 2^k = mu^2\}| \leq 2$ for squarefree odd $m \geq 3$, sharp, via an elementary 2-adic and mod-8 analysis with no equidistribution input) that this range contributes only $O(L)$ to the relevant count, for every n. This range is therefore **closed, unconditionally, for every** n – the only range, besides $p \leq L+1$, with that status.

7.2 Open link (M″): the multiplicity-distribution tail on $(L, L^2]$

Precisely, what is missing to upgrade Lemma 5.2 from a deterministic *reduction* to a deterministic *bound* is:

$$(\mathrm{M}'') \qquad \sum_{p\in(L,L^2]\ \text{non-Sidon}} \big(M_p - 2\big) \;=\; o(L^2), \qquad \text{equivalently} \qquad \sum_{k\geq 2} \#\{p \in (L, L^2] : M_p^{\text{unord}} \geq k\} \;=\; o(L^2).$$

Where brute force fails, exactly. The trivial bound $M_p - 2 \leq L - 1$ together with the trivial bound $\leq \pi(L^2)$ on the number of non-Sidon primes gives $\Theta(L^2)$, not $o(L^2)$ – it fails by a constant factor, not by an order of magnitude. **What the data shows.** M_p^{unord} (the maximum number of *unordered* representations of any residue $n \bmod p^2$ as $\{2^i, 2^j\}$) is *not* bounded by an absolute constant: it grows slowly, empirically $3, 4, 4, 4, 5, 6$ for $L = 20, 30, 40, 50, 60, 70$, attained on primes with anomalously small multiplicative order of 2 (e.g. $p = 127$, $\mathrm{ord}_p(2) = 7$). We checked, and explicitly ruled out, the natural attempt to bound M_p via Evertse's S-unit equation theorem: that theorem bounds non-degenerate solutions of S-unit equations over number fields (characteristic 0); it does not apply to equations in the finite ring $\mathbb{Z}/p^2$, where S-unit-type equations $x + y = z + w$ over a multiplicative subgroup H typically have $\sim |H|^2/p^2 \cdot |H|$ solutions, unbounded as $|H| \to \infty$ – the opposite regime from characteristic 0. The proportion of non-Sidon primes in $(L, L^2]$ is itself a constant fraction ($\approx 17\%$, stable from $L = 32$ to $L = 48$), not a vanishing one – so (M″) is genuinely a statement about the *decay of the multiplicity distribution* (a $B_2[g]$-type statement: how many primes have additive multiplicity $\geq k$, for each k), not merely about how many primes fail to be Sidon. **A cleaner equivalent form.** Since $\#\{\text{non-Sidon } p \in (L, L^2]\} \leq \pi(L^2) = O(L^2/\log L)$ unconditionally, writing the average excess $\mathrm{EM}(L) := \big(\sum_{\text{non-Sidon}}(M_p - 1)\big)/\#\{\text{non-Sidon}\}$ gives the equivalence

$$(\mathrm{M}'') \quad \Longleftrightarrow \quad \mathrm{EM}(L) = o(\log L).$$

Empirically $\mathrm{EM}(L) \approx 1.3$ (computed to $L = 120$: it lies in $[1.27, 1.33]$, with $\mathrm{EM}(L)/\log L$ *decreasing*, $0.33 \to 0.27$), so (M″) holds with a growing margin; correspondingly $\sum_p M_p/L^2$ decreases

($0.16 \to 0.135$). We had conjectured $\mathrm{EM} \to 4/\pi = 1.273$ from data to $L = 80$; extending to $L = 120$ *refutes* a clean closed form (the value drifts up to ≈ 1.30). What is robust is only $\mathrm{EM} = O(1)$, indeed $o(\log L)$. **The shape of the distribution.** The reason $\mathrm{EM} = O(1)$ holds is that the tail counts $\#\{p : M_p^{\mathrm{unord}} \geq k\}$ decay *geometrically* in k: on the Type B primes (§5.3) the ratio $\#\{M_p \geq k{+}1\}/\#\{M_p \geq k\}$ is ≈ 0.08 and stable across $L = 60, \dots, 180$ (n up to $\sim 10^{54}$), with $\#\{M_p \geq 5\} = 0$ throughout; the excess $\sum_k \#\{M_p \geq k\}$ is thus carried almost entirely by the level $k = 3$, and $\#\{M_p \geq 2\} \leq \pi(L^2) = o(L^2)$ is unconditional. We stress that this geometric decay is *observed*, not proved inductively, and that it is the empirical *shape* of the same distribution, not an independent simplification: proving the decay uniformly in k is exactly proving the sub-Poisson behaviour below, and does not sidestep it.

Why four standard toolkits all fail, for one reason. We tried, and ruled out, the four natural routes; each controls an *energy*-scale quantity, whereas (M$''$) needs the *maximum multiplicity*, and for these near-Sidon geometric sets the maximum is far below the energy:

- *Cauchy–Schwarz / additive energy:* $M_p - 1 \leq \sqrt{2\Delta_p}$, but $\sum_p \sqrt{2\Delta_p} = \Theta(L^2)$ (ratio ≈ 0.25, not $\to 0$); and $\sum_p \Delta_p$ itself *grows* faster than L^2.
- *S-unit / Evertse:* inapplicable in the finite ring $\mathbb{Z}/p^2$ (as above; S-unit-type equations there have $\sim |H|^3/p^2$ solutions).
- *Plünnecke–Ruzsa:* the geometric set $G_L = \{2^k \bmod p^2\}$ has *maximal* doubling $|G_L{+}G_L|/|G_L| \sim L$, so Plünnecke gives only the trivial $E \leq K^2|G_L|^2 \sim L^4/4$. (Yet $E/|G_L|^2 \approx 2$–4: G_L is "almost Sidon", energy barely above the floor.)
- *Exponential sums (Bourgain–Garaev–Konyagin–Shparlinski):* a pointwise bound on $S(t) = \sum_{k\leq L} e_{p^2}(t2^k)$ bounds the energy $E_p = p^{-2}\sum_t |S(t)|^4$, not M_p, and $M_p \ll \sqrt{E_p}$ (measured gap $M_{127} = 9$ vs $\sqrt{2\Delta_{127}} = 167$ at $L = 120$). Moreover our sums are incomplete of length $L + 1 \in [q^{1/4}, q^{1/2}]$, $q = p^2$, where even Korobov's $\sqrt{q}\log q$ is trivial – below the range of known unconditional cancellation for the exponential function mod p^2. *A p-adic reduction (Lemma K applied to the moment, not just the main term) partly clarifies where the difficulty sits*: splitting $S(t)$ by whether $p \mid t$, the $p \mid t$ characters $t = pb$ give $S(pb) = \sum_{k\leq L} e_p(b\,2^k)$ – a *complete* sum over $\langle 2\rangle$ mod p when $e_p \leq L$ (Type A, already unconditional by Proposition 5.4), an incomplete geometric sum mod p otherwise – whereas the $p \nmid t$ characters give genuinely incomplete sums mod p^2 that Lemma K does *not* reduce. So the true frontier is not Korobov's generic bound but *sum-product* cancellation for the geometric progression itself (Bourgain–Glibichuk–Konyagin: incomplete geometric sums of length L mod p or p^2, valid for order $e_p \geq p^\delta$); whether such a bound reaches length L in this range, in mean square, is precisely the open input (TB), which we have not settled.

Equivalently (double counting), $\sum_p \Delta_p = \sum_{\text{nontrivial }(a,b,c,d)} \#\{p : p^2 \mid N\}$ with $N = 2^a + 2^b - 2^c - 2^d \neq 0$ (the latter is exactly why $\{2^k\}$ is a Sidon set over $\mathbb{Z}$); here $\#\{p \in (L, L^2] : p^2 \mid N\} \leq 2$ in range, and the only $\omega_2(N) = 2$ cases are the structured squares $N = (2^k - 1)^2$ (e.g. $(2^{14} - 1)^2 = 3^2 \cdot 43^2 \cdot 127^2$). All of this controls the *column/row* marginals (per-quadruplet, per-prime energy) but never the *pile* marginal M_p (e.g. for $p = 127$, $L = 40$: row $\Delta_p = 329$ collisions fan into piles of max $M_p = 3$). So (M$''$) is, precisely, a *large-deviation / multiplicity-distribution* statement about the geometric progression $\{2^k \bmod p^2\}$ – the maximum pile, not the energy – which lies beyond all four *energy*-scale toolkits above. (A fifth, maximum-scale family – Stepanov / García–Voloch – does bound the pile directly (Proposition 5.7), but on Type A alone and short of the target by a power of L.) We regard it as true (every empirical margin grows) but accessible only to a dedicated argument of that kind, which we do not have. As a structural aside, the high-M_p primes correlate strongly with *small* multiplicative order $\operatorname{ord}_p(2)$ ($\operatorname{corr}(M_p, 1/\operatorname{ord}_p(2)) \approx 0.71$ at $L = 60$), though large-order primes also occur among the non-Sidon ones, so this does not by itself reduce the sum.

The refined picture (§5.3–5.4). The correlation with $\operatorname{ord}_p(2)$ is in fact the entry point to a clean split of (M″). Writing Type A $= \{e_p \le L\}$, Type B $= \{e_p > L\}$: the Type A excess is $\le \kappa_p(\lfloor L/e_p \rfloor + 1)$ per prime (Proposition 5.6), and the order sum $\sum_p \lfloor L/e_p \rfloor = o(L^2)$ is now *unconditional* (Proposition 5.4); the only missing Type A input is a bound on the number κ_p of active progressions (hypothesis (TA)), for which a *fifth*, maximum-scale toolkit – the Stepanov / García–Voloch method for additive representations in a multiplicative subgroup – gives the unconditional $\kappa_p \ll e_p^{2/3}$ and hence $\sum_{\text{Type A}} M_p = O(L^{8/3}/\log L)$, short of $o(L^2)$ by $L^{2/3}$ (Proposition 5.7). The Type B excess is governed by the factorial moments E_k^{tot} (Lemma 5.8), and the sharp condition $E_k^{\text{tot}} \le C^k E_k^{\text{null}}$ (hypothesis (TB), Theorem 5.9) reduces it to a Heath-Brown square-sieve estimate, i.e. a *mean-square* bound on the same incomplete exponential sums that appear in sup-norm in the fourth toolkit above. Thus (M″) is not monolithic: its order-scale part is unconditional, its Type A part is a rigidity statement (now with the unconditional partial bound $\kappa_p \ll e_p^{2/3}$ above), and its Type B part is the one genuine analytic input. It is worth stressing how *singular* this lock is: even the unconditional almost-all density of §6 cannot be improved past the first moment without it. The variance of the collision count $C(n) = \sum_p N_p(N_p - 1)$ is $\sum_p \operatorname{Var}(N_p(N_p - 1))$, whose per-prime terms are the sums $\sum_r c_r^4/p^2$ – i.e. the very moments E_4^{tot}; likewise a truncation $C^{\le K} + C^{>K}$ pushes the tail onto $\sum_p \sum_r \binom{c_r}{K}$, again E_K^{tot}. So Chebyshev (numerically a factor-L^2 gain to $O(1/(L^3 \log L))$, $\operatorname{Var}(C)/\overline{C} \approx 2.1$–$2.9$ to $L = 36$) and every higher-moment sharpening are conditional on the *same* square-sieve input as (M″) itself. There is one underlying lock, and it is the multiplicity distribution of $\{2^l + 2^m \bmod p^2\}$.

7.3 Open link (B): the equidistribution wall on $(L^2, \sqrt{n/2}]$ – worst case only

On average this range is already handled (Theorem 6.6: its first moment is $O(1/\log L)$, so it contributes nothing for almost all n). What remains genuinely open is the *uniform-in-n* (worst-case) statement:

$$\text{(B)} \qquad \sum_{p \in (L^2, \sqrt{n/2}]} N_p(n) \;=\; o(L^2) \quad \text{uniformly in } n.$$

Where brute force fails, exactly. For p in this range, $N_p(n) \in \{0, 1\}$ generically (no two pairs can plausibly collide once p is this large relative to the number of pairs), so the sum is, up to a constant, the number of pairs (l, m) for which $n - 2^l - 2^m$ has *some* square prime factor p^2 with $L^2 < p^2 \le n/2$. Each $k = n - 2^l - 2^m \le n$ has at most $O(L/\log L)$ such prime factors (since $(L^2)^{L/\log L} \sim n$), so the trivial bound is $(L+1)^2 \cdot O(L/\log L) = O(L^3/\log L)$ – the *cube*, not the square, of L: this range is, in the crudest sense, the *numerically dominant* one (most primes up to $\sqrt{n}$ lie here), even though the data shows its actual contribution is small. **What would close it.** A squarefree-density / equidistribution estimate for the structured sequence $\{n - 2^l - 2^m\}$ in this range – in spirit close to (but not identical to) the kind of input behind the classical "almost all n" result for the one-power variant of #11 itself. The covering-congruence machinery that produces the large exceptional sets for the *prime*-kernel problems #9 and #16 (Crocker [3], Pan [5], Chen [4]) is *provably unavailable* here: a covering system needs its moduli to satisfy $\sum 1/m_i \ge 1$, whereas square moduli p^2 ($p \ge 3$) give only $\sum_{p \ge 3} 1/p^2 = 0.2022\ldots < 1$, so one cannot cover $\mathbb{Z}$ by congruences mod p^2. This is the quantitative form of the Crocker decoupling of §1.3 – and the same smallness ($\sum_{p > L^2} 1/p^2$ tiny) is exactly what makes the average bound Theorem 6.6 work.

The worst case (B) would, by contrast, require a genuine equidistribution/large-sieve estimate for the structured sequence $\{n - 2^l - 2^m\}$; we have not found one.

We note that (WB) is of the *same nature* as, and in fact weaker than, the Type B hypothesis (TB) of §5.4: both are factorial-moment (grand sieve) bounds on the multiplicity distribution of $\{2^l + 2^m \bmod p^2\}$, (WB) over the larger primes $p > L^2$ (where collisions are rarer, so the maximum $\max_n B(n)$ is smaller). The naive hope that $\max_n B(n) = O(1)$ is *false*: $p \le \sqrt{n/2} \le 2^{L/2}$ gives $p^2 \le 2^L$, while a difference of two pair-sums has absolute value up to $4 \cdot 2^L > p^2$, so the "at most one pair per fibre" argument does not apply in this range; numerically $\max_n B(n)$ tracks the Poisson maximum $\approx 0.69\, L/\log L$ (not $O(1)$), which is nonetheless $o(L^2)$, so (WB) holds with a wide margin. Thus (TA), (TB) and (WB) are three scales of a single underlying question – the multiplicity distribution of $\{2^l + 2^m \bmod p^2\}$ – rather than three independent walls.

7.4 Summary table

Range of p	Status (every n)	Method
$p \le L + 1$	**closed**	Lemma K + Prop. 4.2 (elementa
$(L, L^2]$, Type A ($e_p \le L$)	*open*, gap $L^{2/3}$	$O(L^{8/3}/\log L)$, García–Voloch (
$(L, L^2]$, Type B ($e_p > L$)	*open* ((TB); sup-norm route closed by circularity)	(TB) = additive energy (Prop. 5
$(\sqrt{n/2}, \sqrt{n}]$	**closed**	Kalinin [2]; a.a. n self-contained
$(L^2, \sqrt{n/2}]$	*open*, worst case only	wall (B); #11-calibre (avg. done

(All four ranges are **closed unconditionally for almost all** n, Corollary 6.9.)

8 The function-field model: where the difficulty really lives

To locate the obstruction precisely we run the whole architecture over $\mathbb{F}_q[t]$. Replace $\mathbb{Z}$ by $\mathbb{F}_q[t]$, the base 2 by the variable t, primes p by monic irreducibles P, $|p|$ by $|P| = q^{\deg P}$, and "squarefree" by squarefree in $\mathbb{F}_q[t]$ (density $1 - 1/q$). The variant reads: every $n(t)$ of large degree is $k(t) + t^l + t^m$ with k squarefree. The sieve is identical, with $N_P(n) = \#\{(l, m) : P^2 \mid n - t^l - t^m\}$ and threshold $T = (L + 1)^2$. Lemma K transfers verbatim (the kernel $K_P = 1 + P\,\mathbb{F}_q[t]/P^2$ of $(\mathbb{F}_q[t]/P^2)^\times \to (\mathbb{F}_q[t]/P)^\times$ absorbs the character sum), as does the involution $1+t^{e-\delta} \equiv t^{-\delta}(1+t^\delta)$; and $\sum_P |P|^{-2} = \sum_d (\#\{\deg P = d\}) q^{-2d} < 1$ (for $q = 2$, ≈ 0.60), so covering by congruences mod P^2 is impossible – the Crocker decoupling holds here too.

The decisive difference is at (TB).

Proposition 8.1 ((TB) over $\mathbb{F}_q[t]$, by degree)**.** *Let $q > 2k$. For an irreducible P with $\deg P > L/2$, the sums $\{t^l + t^m \bmod P^2\}$ have no non-trivial coincidence, so $N_P(r) \le 1$ for the relevant residues and the k-th factorial moment vanishes. Consequently the $\mathbb{F}_q[t]$-analogue of (TB) holds unconditionally.*

Proof. A coincidence $\sum_i t^{a_i} \equiv \sum_i t^{b_i} \pmod{P^2}$ means P^2 divides the polynomial $D = \sum_i t^{a_i} - \sum_i t^{b_i}$, of degree $\le L$ and with integer coefficients reduced mod q. As $q > 2k$, D vanishes in $\mathbb{F}_q[t]$ iff the exponent multisets coincide. If they do not, $D \ne 0$ has $\deg D \le L < 2 \deg P = \deg P^2$, so $P^2 \nmid D$. Hence the only coincidences are trivial. For $\deg P \le L/2$ (finitely many P, $|P| \le q^{L/2}$) the coincidences are counted by degree, without any exponential-sum input. □

So over $\mathbb{F}_q[t]$ the analogue of (TB) is a *theorem*, but proved by the linear independence of the monomials t^j and a degree count – not by Weil, and not by any sup-norm bound. Indeed the moment/sup-norm obstruction of §5.5 persists here: Weil gives square-root cancellation $\varepsilon = \frac{1}{2}$, still $B^2 = (L+1) > 1$, so the conversion of Proposition 5.12 blocks identically; the closure comes from elsewhere. And the mechanism does *not* transpose to $\mathbb{Z}$: there is no degree bound preventing $p^2 \mid 2^a + 2^b - 2^c - 2^d \neq 0$. The obstruction over $\mathbb{Z}$ is thus *archimedean* – powers of 2 carry (e.g. $2^a + 2^a = 2^{a+1}$) while monomials t^a do not, so $\{2^j\}$ is only *approximately* Sidon and a nonzero four-term combination can still be squarefull – rather than a deficit of cancellation. The function-field model therefore supplies no characteristic-zero method; it certifies that (TB) over $\mathbb{Z}$ is genuinely harder, of square-sieve / squarefull-value type.

9 Conclusion

The two-powers squarefree variant of Erdős #11 is, as far as we have been able to push it, reduced to exactly two clean, isolated, named statements – (M$''$), a multiplicity-distribution decay on a bounded range of primes just above $L = \log_2 n$, and (B), a squarefree-density statement on the much wider range of primes between L^2 and $\sqrt{n}$ – after an elementary closure (Lemma K and Proposition R1) that handles, exactly and for every n, every prime up to $L+1$, and a deterministic dissolution of the worst-case obstruction (Lemma M) on the dominant range $(L, L^2]$. An unconditional theorem (§6–6.2) further proves the variant for *almost all* n, on *all* prime ranges (Corollary 6.9).

Within (M$''$) the split of §5.3–5.4 is now sharp. The order-scale part is unconditional (Proposition 5.4). The Type A part carries a genuine *unconditional* bound $\kappa_p \ll e_p^{2/3}$ via García–Voloch (Proposition 5.7), giving $\sum_{\text{Type A}} M_p = O(L^{8/3}/\log L)$ – short of the target $o(L^2)$ by exactly $L^{2/3}$, so (TA) survives only as the sharp question of pushing $2/3 \to o(1)$ using the involution rigidity. The Type B part (TB) is where the problem's true nature surfaces. The relevant *additive* exponential sums do satisfy an unconditional sup-norm bound $|S(\xi)| < (L+1)^{1-\varepsilon}$ (Bourgain–Chang, Theorem 5.11, confirmed numerically), yet this cannot close (TB): the moment (L^{2k}) bound one would need is, by Proposition 5.13, *identical* to the additive energy E_k^{add}, i.e. to (TB) itself. (TB) is therefore closed by *circularity*, not by any absent reference – it is a genuine open estimate. The function-field model (§8) pins down why: the analogue of (TB) over $\mathbb{F}_q[t]$ is a theorem, but proved by the linear independence of the monomials t^j and a degree count, a mechanism that does *not* transpose because over $\mathbb{Z}$ nothing bounds the squarefull part of a nonzero four-term combination $2^a + 2^b - 2^c - 2^d$. The obstruction is *archimedean* – powers of 2 carry while monomials do not – not a deficit of cancellation. In this precise sense (M$''$) at $k=2$ is a squarefree-value statement for the sparse family $1 + 2^\beta - 2^\gamma - 2^\delta$, a genuine cousin of #11 itself; empirically the few quadruples that a single p^2 divides twice are *all* structured (Mersenne squares $N = (2^k-1)^2$ and relatives; §7, script data), suggesting an elementary classification for $k=2$ that we state as a conjecture, not a theorem. As for (B): its average form is settled (first moment); its worst case remains, and we judge it to be of the same calibre as #11 in its original one-power form. The *elementary* lemmas underpinning the unconditional results (Proposition 5.4, the order-sum bound; Lemma 5.5, the involution identity; the order lower bound behind Proposition 2.4; the coprimality product bound behind Proposition 5.4; and the degree obstruction behind the function-field Proposition 8.1) have been *formally verified in Lean 4* against Mathlib (file `Erdos11_verified.lean`, five lemmas, `0 sorry`, axiom audit `[propext, Classical.choice, Quot.sound]`); the analytic and conditional statements (the factorial-moment bound (TB), the every-n theorem) are *not* formalised, matching their status as conjectural. Only the target statement `erdos_11.variants.two_pow_two` otherwise exists in Lean form. We hope the precision of the named gaps is useful to whoever picks this up next.

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This note was produced in a human-directed workflow with AI assistants in distinct executor and validator roles: proposing tools and computations, locating and cross-checking references, and deriving and refuting bounds, with a standing rule that no theorem enters the paper without a written proof or a formal (Lean) verification, and every numerical assertion points to a named script (see the repository `README`). The mathematical judgements, and the responsibility for errors, are the author's.

References

[1] T. F. Bloom, *Erdős Problem #11*, https://www.erdosproblems.com/11.

[2] D. Kalinin, *A multiplicity bound for a fixed squarefree kernel*, preprint, 29 March 2026. We rely on this paper for the exact statements of, and exact references to, Granville–Soundararajan, Hercher, Crocker and Platt–Trudgian on the one-power variant of #11, and for Theorem 4 used in §7.

[3] R. Crocker, *On the sum of a prime and two powers of two*, Pacific J. Math. **36** (1971), 103–107. (Title and pages confirmed via the bibliography of [4] and [5].)

[4] Y.-G. Chen, *A conjecture of Erdős on $p + 2^k$*, arXiv:2312.04120v3 (2024) – the resolution of Erdős Problem #16 (the *one*-power exceptional-set structure), used here only to fix the exact references and the covering-congruence mechanism discussed in §7.

[5] H. Pan, *On the integers not of the form $p + 2^a + 2^b$*, Acta Arith. **148** (2011), 55–61 (arXiv:0905.3809) – the *prime*-kernel two-powers problem (Erdős #9); its lower bound on non-representable integers is obtained by a covering system modulo primes, not a sieve.

[6] D. R. Heath-Brown, *The square sieve and consecutive square-free numbers*, Math. Ann. **266** (1984), 251–259.

[7] A. García, J. F. Voloch, *Fermat curves over finite fields*, J. Number Theory **30** (1988), 345–356 (primary source of the multiplicity bound $r_p(b) \le 4|G|^{2/3}$).

[8] S. V. Konyagin, *Estimates for character sums and the additive structure of multiplicative subgroups* (lecture notes; recordings and notes available via https://www.mathtube.org/); and S. V. Konyagin, I. E. Shparlinski, *Character sums with exponential functions and their applications*, Cambridge Univ. Press, 1999. We take the exact form used from here (Konyagin's Theorem 2.1): $r_p(b) \le 4|G|^{2/3}$ for $b \in \mathbb{Z}_p^\times$ and $|G| < (p-1)/((p-1)^{1/4}+1)$; the Stepanov-method proof (Heath-Brown) and the energy bound $T_2(G) \ll |G|^{5/2}$ for $|G| \le p^{2/3}$ (Heath-Brown–Konyagin) are stated there. The exact size and non-degeneracy conditions used in Proposition 5.7 are those of this secondary source, verified numerically herein.

[9] J. Bourgain, M.-C. Chang, *Exponential sum estimates over subgroups and almost subgroups of $\mathbb{Z}_Q^*$, where Q is composite with few prime factors*, Geom. Funct. Anal. **16** (2006), 327–366. Corollary 4.5 gives the additive sup-norm of Theorem 5.11; formula (4.23) ($\theta = 1 + p$) is the degeneracy that forces the Type A/B split.

[10] W. Rudin, *Trigonometric series with gaps*, J. Math. Mech. **9** (1960), 203–227.

[11] Numerical evidence referenced throughout (scripts `orders.py`, `lemmaE2.py`, `verify_structure.py`, `verify_residual.py`, and further task-specific computations for M_p, $C(n)$, and the non-Sidon-prime counts, all in this repository).